# Software Requirements Specification (SRS)

**Project:** Digital Campus Library Reservation Gateway
**Version:** 1.0
**Authors:** Dhareen G (SRN: PES1UG24AM084)
**Date:** 28-10-2026
**Status:** Draft for submission

## Revision history

| Version | Date | Author | Change summary | Approval |
|---|---|---|---|---|
| 1.0 | 28-10-2026 | Dhareen G | First complete draft | Pending |

## Approvals

| Role | Name | Signature / Email | Date |
|---|---|---|---|
| Course Coordinator | | | |
| Project Guide | | | |

## Table of Contents

1. Introduction
2. Overall description
3. External interface requirements
4. System features (detailed)
5. Non-functional requirements (detailed)  (5.1 Security)
6. Quality attributes and acceptance tests
7. System models and diagrams (UML use-case diagrams)
8. Requirements Traceability Matrix (RTM)

# 1. Introduction

## 1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Digital Campus Library Reservation Gateway (LRG). It defines the functional, non-functional and security requirements, the external interfaces and the verification criteria for version 1.0, for use by developers, testers and evaluators. Companion documents are the Software Architecture and Design Specification (SAD) and the Software Test Plan (STP).

## 1.2 Scope

LRG is a web portal through which student members search the campus library catalog by ISBN, title or author, see whether a title is Available or CheckedOut, and place and manage holds on checked-out titles. The Head Librarian keeps availability and catalog data current and monitors hold demand. The system notifies the next member in a queue by email when a title is returned. Out of scope: physical check-out and check-in hardware, fines and payments, inter-library loans, and the internals of the campus SSO and mail services.

## 1.3 Audience

Developers, QA engineers, the Head Librarian and library IT staff, and course evaluators.

## 1.4 Definitions

| Term | Meaning |
|---|---|
| LRG | Library Reservation Gateway (this system) |
| ISBN | International Standard Book Number (10 or 13 digits) |
| Hold | A reservation by a member on a CheckedOut title, kept in a first-come first-served queue |
| Ready | Hold state meaning the title has been returned and is kept for the first queued member for 72 hours |
| SSO | Single sign-on: the campus identity provider used for login |
| RBAC | Role-based access control |
| TLS | Transport Layer Security |
| API | Application Programming Interface |
| RTM | Requirements Traceability Matrix |
| WCAG | Web Content Accessibility Guidelines |
| UC | Use case |

# 2. Overall description

## 2.1 Product perspective

LRG is a new, self-contained web application with a layered architecture (presentation, business logic, data). It depends on two external services: the campus SSO provider for authentication and an SMTP email service for notifications. It owns a relational Library Database holding titles, members' hold records and audit logs.

## 2.2 Major product functions

- Search the catalog by ISBN, title or author with paged results and availability status (UC-01, UC-02)
- Place, view and cancel holds; see queue position (UC-03, UC-04)
- Log in through campus SSO; guests may search only (UC-05)
- Update availability on return or checkout and promote the next hold (UC-06)
- Notify the next member by email (UC-07)
- Manage catalog entries and view a hold-demand report (UC-08, UC-09)

### 2.3 User roles and characteristics

| Role | Characteristics | Main use cases |
|---|---|---|
| Guest | Not logged in; can only search | UC-01, UC-02 |
| Student Member | Campus student; basic web skills; expects fast, clear answers; typically on a phone or laptop | UC-01 to UC-05 |
| Head Librarian | Library staff; administers availability, catalog and demand data | UC-01, UC-05 to UC-09 |
| Campus SSO (external) | Identity provider that authenticates users | UC-05 |
| Email Service (external) | SMTP service that delivers notifications | UC-07 |

### 2.4 Operating environment

Server: Linux host or cloud VM running the application server and a relational database. Clients: current Chrome, Firefox and Edge, desktop and mobile. Network: campus network and internet, HTTPS only.

### 2.5 Constraints

- Authentication must use the campus SSO; LRG must not store passwords.
- Member data is personal data and must be protected under the institution's data-protection policy.
- Project delivered in a one-semester course timeline using open-source tools.
- Emails are sent through the institution's SMTP service; delivery delays are outside LRG's control.

### 2.6 Assumptions

Title availability is maintained by the Head Librarian in LRG; there is no live link to a check-out desk system in version 1.0. Every member has a campus email address.

## 3. External interface requirements

### 3.1 User interfaces

A responsive web UI with a search bar on the home page, a paginated result list, a book detail page, a My Holds page, and a librarian dashboard visible only to the Head Librarian. Accessibility: keyboard navigation, labelled form fields, sufficient colour contrast (WCAG 2.1 AA).

### 3.2 Hardware interfaces

None. The system runs on standard server hardware and is accessed from standard client devices.

### 3.3 Software interfaces

- **Campus SSO:** OpenID Connect / SAML login; returns a signed token with member id, name, email and role.
- **Email service:** SMTP over TLS for hold-ready and reminder emails.
- **Library Database:** relational database accessed only through the data layer using parameterized queries.

### 3.4 Communications

HTTPS with TLS 1.2 or higher for all traffic. JSON over REST between the portal front end and the application API. Failed emails are retried up to three times with back-off and then logged.

## 4. System features (detailed)

Each requirement lists priority, source, acceptance criteria and a test-case reference. Functional requirement IDs follow LRG-F-###.

## 4.1 Catalog Search (UC-01)

**Description:** Search the catalog by ISBN, title or author, view availability and open book details.

| Req ID | Requirement (shall...) | Type | Priority | Source / Stakeholder | Acceptance criteria / Test case ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| LRG-F-001 | The system shall allow a user to search the catalog by ISBN, title or author. | Functional | High | Student Member / Head Librarian | AC-LRG-F-001: A valid ISBN returns the matching title; title and author searches return all partial matches. Test: TC-SRCH-01, TC-SRCH-02 | Library Database |
| LRG-F-002 | The system shall validate that the search term is non-empty and properly formatted (ISBN-10/13 syntax when an ISBN is entered) before querying the catalog. | Functional | High | Student Member | AC-LRG-F-002: Empty or malformed input is rejected with a message and no database query runs. Test: TC-SRCH-03 | Library Database |
| LRG-F-003 | The system shall return results in pages showing title, author, ISBN and availability status (Available / CheckedOut). | Functional | High | Student Member | AC-LRG-F-003: Each page of 10 results shows all four fields; next and previous page controls work. Test: TC-SRCH-04 | Library Database |
| LRG-F-004 | The system shall display a book detail page and show a Place Hold option only when the title is CheckedOut. | Functional | Medium | Student Member | AC-LRG-F-004: The detail page lists full metadata; Place Hold appears only for CheckedOut titles. Test: TC-SRCH-05 | Library Database |
| LRG-F-005 | The system shall display "No results found" with advice to check spelling or broaden the search when a query matches no titles. | Functional | Low | Student Member | AC-LRG-F-005: Zero matches shows the message and no result list. Test: TC-SRCH-06 | Library Database |

## 4.2 Authentication and Access Control

**Description:** Campus login, guest search, role separation and session handling.

| Req ID | Requirement (shall...) | Type | Priority | Source / Stakeholder | Acceptance criteria / Test case ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| LRG-F-010 | The system shall authenticate members through campus credentials (single sign-on). | Functional | High | Student Member / Library IT | AC-LRG-F-010: Valid credentials create a session; invalid credentials show an error and create no session. Test: TC-AUTH-01, TC-AUTH-02 | Campus SSO |
| LRG-F-011 | The system shall allow guests to search the catalog without logging in and shall require login before a hold is placed. | Functional | Medium | Student Member | AC-LRG-F-011: A guest can search; selecting Place Hold redirects to login. Test: TC-AUTH-03 | Campus SSO |
| LRG-F-012 | The system shall restrict librarian functions to users holding the Head Librarian role. | Functional | High | Head Librarian / Library IT | AC-LRG-F-012: A Student Member calling any librarian function receives an access-denied response. Test: TC-AUTH-04 | Campus SSO |
| LRG-F-013 | The system shall let a user log out and shall expire a session after 30 minutes of inactivity. | Functional | Medium | Library IT | AC-LRG-F-013: After logout or 30 idle minutes, protected pages require login again. Test: TC-AUTH-05 | Campus SSO |

## 4.3 Hold and Reservation Management

**Description:** Place, view and cancel holds and see queue position, with per-member limits.

| Req ID | Requirement (shall...) | Type | Priority | Source / Stakeholder | Acceptance criteria / Test case ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| LRG-F-020 | The system shall allow a logged-in member to place a hold on a CheckedOut title. | Functional | High | Student Member | AC-LRG-F-020: A hold is created, appended to the title's queue and confirmed with its queue position. Test: TC-HOLD-01 | Library Database, AUTH |
| LRG-F-021 | The system shall reject a hold on an Available title and a second active hold by the same member on the same title. | Functional | High | Head Librarian | AC-LRG-F-021: Both cases return a clear rejection and create no hold record. Test: TC-HOLD-02 | Library Database, AUTH |

| Req ID | Requirement (shall...) | Type | Priority | Source / Stakeholder | Acceptance criteria / Test case ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| LRG-F-022 | The system shall let a member view and cancel their active holds. | Functional | Medium | Student Member | AC-LRG-F-022: My Holds lists active holds; cancelling removes the hold and updates the queue. Test: TC-HOLD-03 | Library Database, AUTH |
| LRG-F-023 | The system shall show a member their position in a title's hold queue. | Functional | Low | Student Member | AC-LRG-F-023: The displayed position is correct and updates after earlier holds are cancelled. Test: TC-HOLD-04 | Library Database, AUTH |
| LRG-F-024 | The system shall limit each member to 5 active holds. | Functional | Medium | Head Librarian | AC-LRG-F-024: A sixth hold request is rejected with an explanatory message. Test: TC-HOLD-05 | Library Database, AUTH |

## 4.4 Notifications

**Description:** Email notices when a held title is ready and before a hold expires.

| Req ID | Requirement (shall...) | Type | Priority | Source / Stakeholder | Acceptance criteria / Test case ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| LRG-F-030 | The system shall notify a member by email when a title they hold becomes available. | Functional | Medium | Student Member | AC-LRG-F-030: When the first queued hold becomes Ready, an email naming the title is sent to that member. Test: TC-NOTF-01 | Email service, RES |
| LRG-F-031 | The system shall send a reminder 24 hours before a Ready hold expires. | Functional | Low | Student Member | AC-LRG-F-031: A reminder email is sent 24 hours before expiry and not after cancellation. Test: TC-NOTF-02 | Email service, RES |

## 4.5 Librarian Administration

**Description:** Keep availability and catalog data correct and monitor hold demand.

| Req ID | Requirement (shall...) | Type | Priority | Source / Stakeholder | Acceptance criteria / Test case ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| LRG-F-040 | The system shall let the Head Librarian update a title's availability status on return or checkout, and shall move the first queued hold to Ready on return. | Functional | High | Head Librarian | AC-LRG-F-040: Marking a title returned sets it Available and, if a queue exists, sets the first hold Ready. Test: TC-ADM-01 | Library Database, AUTH |
| LRG-F-041 | The system shall show the Head Librarian a report of titles ranked by hold-queue length. | Functional | Low | Head Librarian | AC-LRG-F-041: The report lists titles in descending queue length with counts matching the database. Test: TC-ADM-02 | Library Database, AUTH |
| LRG-F-042 | The system shall let the Head Librarian add and edit catalog entries. | Functional | Medium | Head Librarian | AC-LRG-F-042: A new or edited entry is saved and appears in search results. Test: TC-ADM-03 | Library Database |

# 5. Non-functional requirements (detailed)

NFRs are measurable and tied to tests. IDs follow LRG-NF-###.

| Req ID | Requirement | Category | Priority | Acceptance criteria / Measurement |
|---|---|---|---|---|
| LRG-NF-001 | A catalog search shall return its first page within 2 seconds for 90% of requests with 200 concurrent users. | Performance | High | 90th percentile <= 2 s in a JMeter load run. Test: TC-PERF-01 |
| LRG-NF-002 | The system shall be available 99.5% of each month during the academic term, excluding scheduled maintenance. | Reliability | High | Monitoring reports show >= 99.5% monthly uptime. Test: TC-AVL-01 |
| LRG-NF-003 | Member identity and hold history shall be visible only to the member concerned and the Head Librarian. | Security / Privacy | High | No API response contains another member's data. Test: TC-SEC-04 |

| Req ID | Requirement | Category | Priority | Acceptance criteria / Measurement |
|---|---|---|---|---|
| LRG-NF-004 | The portal shall meet WCAG 2.1 AA and work on current Chrome, Firefox and Edge and on mobile browsers. | Usability / Accessibility | Medium | Accessibility audit passes; layout usable at 360 px width. Test: TC-UX-01 |
| LRG-NF-005 | The system shall support 5,000 registered members and 100,000 catalog titles without exceeding the LRG-NF-001 search time. | Scalability | Medium | Load test with seeded data meets LRG-NF-001. Test: TC-PERF-02 |
| LRG-NF-006 | Components shall communicate only through documented interfaces, and business-layer code shall have at least 70% unit-test coverage. | Maintainability | Medium | Coverage report >= 70%; no direct UI-to-database calls. Test: TC-MNT-01 |
| LRG-NF-007 | Hold creation, hold cancellation and librarian status changes shall be logged with user, time and action and retained for 1 year. | Audit / Data integrity | High | Every such action has a log entry retrievable for 1 year. Test: TC-AUD-01 |

## 5.1 Security

### 5.1.1 Security objectives

| Objective | Description |
|---|---|
| SO-1 Confidentiality | Protect member identity, login data and hold history from unauthorized disclosure. |
| SO-2 Integrity | Ensure availability status and hold queues change only through authorized, validated and auditable actions. |
| SO-3 Availability | Keep search and hold services available and resistant to abuse such as brute-force login, scraping and request flooding. |

### 5.1.2 Security requirements

| Req ID | Requirement (shall...) | Type | Priority | Acceptance criteria / Test case ref |
|---|---|---|---|---|
| LRG-SR-001 | TLS 1.2 or higher shall be mandatory for all connections between browser, application, SSO provider and mail server. | Security | High | A scan shows only TLS 1.2/1.3; plain HTTP redirects to HTTPS. Test: TC-SEC-01 |
| LRG-SR-002 | The system shall never store or log member passwords; authentication shall rely on campus SSO tokens. | Security | High | Database and logs contain no passwords. Test: TC-SEC-05 |
| LRG-SR-003 | Every hold and librarian operation shall be authorized on the server by role and ownership. | Security | High | Direct API calls with a Student token to librarian endpoints, or to another member's hold, return 403. Tests: TC-AUTH-04, TC-SEC-06 |
| LRG-SR-004 | All user input shall be validated and all database access shall use parameterized queries. | Security | High | SQL injection and script payloads are rejected or neutralized. Test: TC-SEC-02 |
| LRG-SR-005 | After 5 failed logins within 15 minutes, further attempts from that account or address shall be blocked for 15 minutes and the event logged. | Security | High | Sixth attempt is blocked and logged. Test: TC-AUTH-02 |
| LRG-SR-006 | Session cookies shall be marked Secure, HttpOnly and SameSite=Strict. | Security | Medium | Response headers show all three attributes. Test: TC-SEC-03 |
| LRG-SR-007 | Search and API endpoints shall be rate-limited to 60 requests per minute per client. | Security | Medium | Request 61 within a minute returns HTTP 429. Test: TC-SEC-07 |

# 6. Quality attributes and acceptance tests

- Exit criteria for acceptance: all High-priority functional requirements implemented and verified, no critical NFR or security failure, and the RTM shows every test case passed.
- Acceptance test suites: Catalog Search, Authentication, Holds, Notifications, Librarian Administration, Performance, Security, Accessibility.
- Quality attributes in priority order: security and privacy, correctness of availability and hold queues, performance, usability and accessibility, maintainability.
- Detailed test cases are listed in the Software Test Plan, Appendix A.

# 7. System models and diagrams

## 7.1 UML use-case diagrams

Figure 1 covers the member-facing use cases and Figure 2 covers librarian administration. Use cases and their primary actors:

| Use case | Name | Primary actor | Related requirements |
|---|---|---|---|
| UC-01 | Search ISBN Catalog | Student Member, Head Librarian, Guest | LRG-F-001 to 005 |
| UC-02 | View Book Details | Student Member, Guest | LRG-F-004 |
| UC-03 | Place Hold | Student Member | LRG-F-020, 021, 024 |
| UC-04 | View and Cancel My Holds | Student Member | LRG-F-022, 023 |
| UC-05 | Log In | Student Member, Head Librarian (Campus SSO) | LRG-F-010 to 013 |
| UC-06 | Update Availability Status | Head Librarian | LRG-F-040 |
| UC-07 | Notify Next Member | Email Service (system-triggered) | LRG-F-030, 031 |
| UC-08 | Manage Catalog Entries | Head Librarian | LRG-F-042 |
| UC-09 | View Hold Demand Report | Head Librarian | LRG-F-041 |

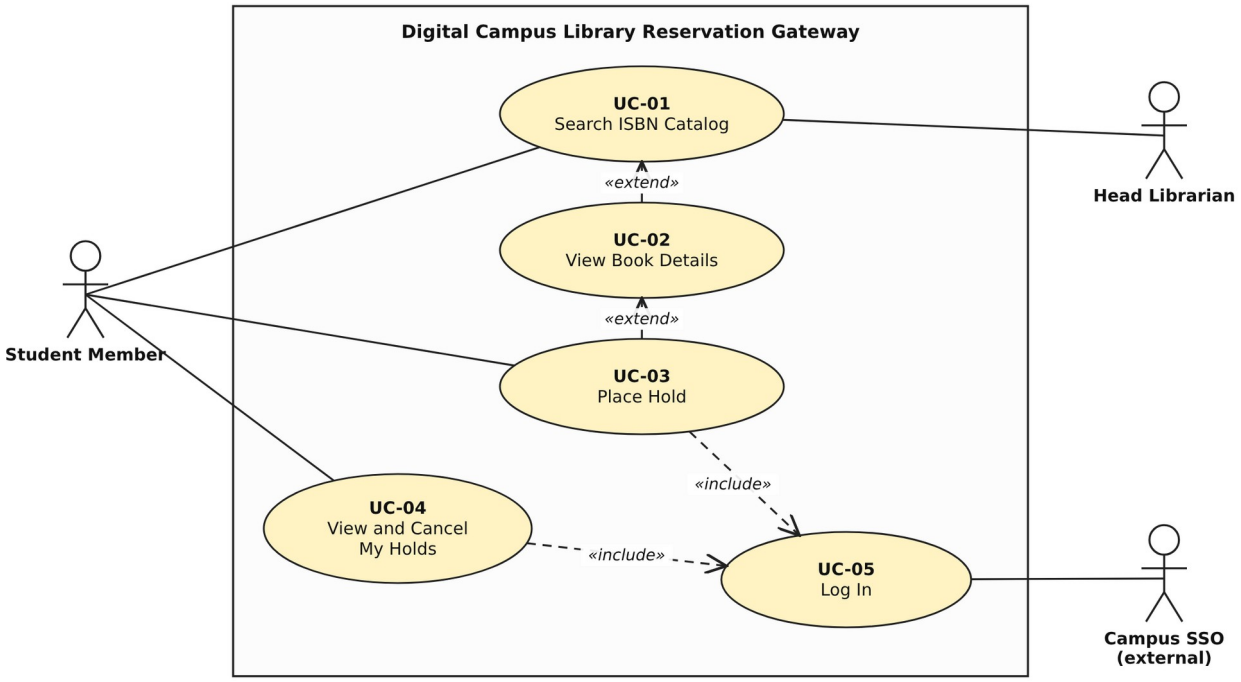



*Figure 1: Use-case diagram 1 - member-facing use cases*

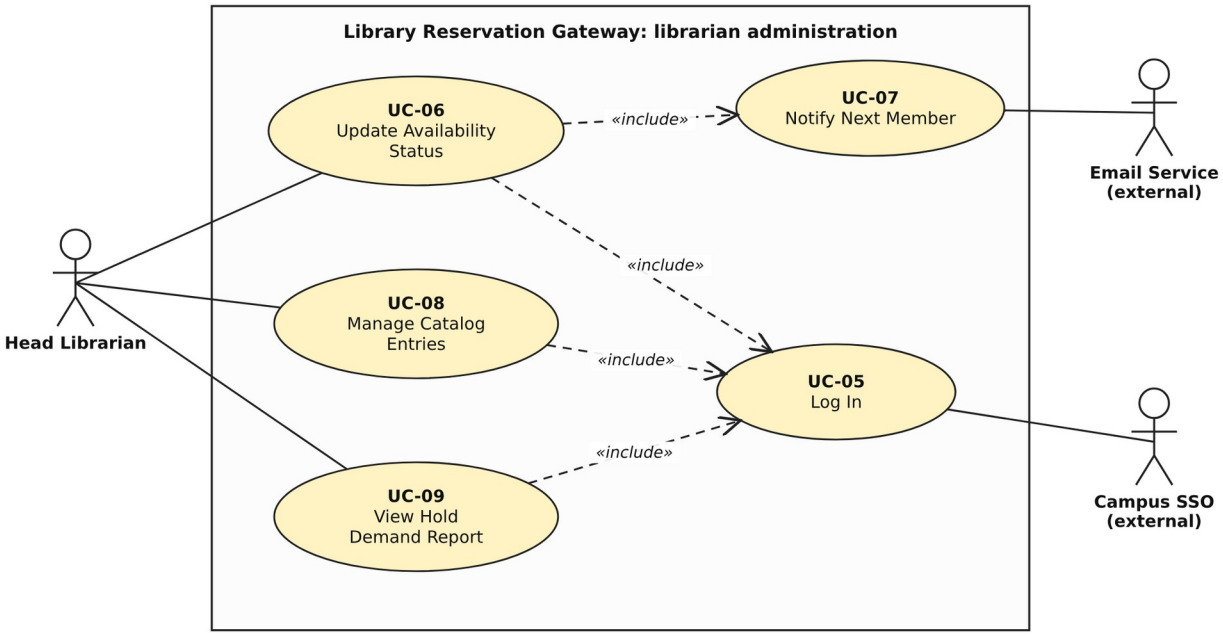



*Figure 2: Use-case diagram 2 - librarian administration*

## 7.2 UC-01 Search ISBN Catalog (specification)

| Item | Description |
|---|---|
| Primary actor | Student Member (Head Librarian and Guest may also search) |
| Precondition | Portal is reachable; catalog contains titles |
| Main flow | 1. Actor enters an ISBN, title or author. 2. System validates the input. 3. System queries the catalog. 4. System shows a page of results with title, author, ISBN and status. 5. Actor selects a title to see its details. |
| Alternate flows | A1: Input empty or malformed - show validation message, no query. A2: No matches - show "No results found" with advice. |
| Postcondition | Results or a message are displayed; no data is changed |
| Requirements | LRG-F-001 to LRG-F-005, LRG-NF-001 |

# 8. Requirements Traceability Matrix (RTM)

Status: N = Not started, P = In progress, A = Accepted. Design references point to the SAD.

| Req ID | Requirement (short) | Section / Design Spec | Module | Test case(s) | Status | Comments |
|---|---|---|---|---|---|---|
| LRG-F-001 | Search catalog | 4.1 / DS-SRCH-01 | Catalog Search | TC-SRCH-01, TC-SRCH-02 | N | |
| LRG-F-002 | Validate search input | 4.1 / DS-SRCH-01 | Catalog Search | TC-SRCH-03 | N | |
| LRG-F-003 | Paged results + status | 4.1 / DS-SRCH-01 | Catalog Search | TC-SRCH-04 | N | |
| LRG-F-004 | Book detail page | 4.1 / DS-SRCH-01 | Catalog Search | TC-SRCH-05 | N | |
| LRG-F-005 | No-results message | 4.1 / DS-SRCH-01 | Catalog Search | TC-SRCH-06 | N | |
| LRG-F-010 | Campus login | 4.2 / DS-AUTH-01 | Authentication | TC-AUTH-01, TC-AUTH-02 | N | |
| LRG-F-011 | Guest search | 4.2 / DS-AUTH-01 | Authentication | TC-AUTH-03 | N | |
| LRG-F-012 | Librarian role | 4.2 / DS-AUTH-01 | Authentication | TC-AUTH-04 | N | |
| LRG-F-013 | Logout / idle expiry | 4.2 / DS-AUTH-01 | Authentication | TC-AUTH-05 | N | |
| LRG-F-020 | Place hold | 4.3 / DS-RES-01 | Reservation Manager | TC-HOLD-01 | N | |
| LRG-F-021 | Reject invalid holds | 4.3 / DS-RES-01 | Reservation Manager | TC-HOLD-02 | N | |
| LRG-F-022 | View / cancel holds | 4.3 / DS-RES-01 | Reservation Manager | TC-HOLD-03 | N | |
| LRG-F-023 | Queue position | 4.3 / DS-RES-01 | Reservation Manager | TC-HOLD-04 | N | |
| LRG-F-024 | 5-hold limit | 4.3 / DS-RES-01 | Reservation Manager | TC-HOLD-05 | N | |

| Req ID | Requirement (short) | Section / Design Spec | Module | Test case(s) | Status | Comments |
|---|---|---|---|---|---|---|
| LRG-F-030 | Availability email | 4.4 / DS-NOT-01 | Notification | TC-NOTF-01 | N | |
| LRG-F-031 | Expiry reminder | 4.4 / DS-NOT-01 | Notification | TC-NOTF-02 | N | |
| LRG-F-040 | Update availability | 4.5 / DS-ADM-01 | Reservation Manager | TC-ADM-01 | N | |
| LRG-F-041 | Hold demand report | 4.5 / DS-ADM-01 | Reservation Manager | TC-ADM-02 | N | |
| LRG-F-042 | Manage catalog entries | 4.5 / DS-ADM-01 | Catalog Search | TC-ADM-03 | N | |
| LRG-NF-001 | Search response time | 5 / DS-PERF-01 | SRCH / DB | TC-PERF-01 | N | |
| LRG-NF-002 | Availability 99.5% | 5 / DS-DEPLOY-01 | All | TC-AVL-01 | N | |
| LRG-NF-003 | Data confidentiality | 5 / DS-SEC-01 | RES / AUTH | TC-SEC-04 | N | |
| LRG-NF-004 | Accessibility | 5 / DS-UX-01 | UI | TC-UX-01 | N | |
| LRG-NF-005 | Scalability | 5 / DS-PERF-01 | SRCH / DB | TC-PERF-02 | N | |
| LRG-NF-006 | Maintainability | 5 / DS-ARCH-01 | All | TC-MNT-01 | N | |
| LRG-NF-007 | Audit logging | 5 / DS-LOG-01 | RES / DB | TC-AUD-01 | N | |
| LRG-SR-001 | TLS 1.2+ | 5.1 / DS-SEC-01 | All | TC-SEC-01 | N | |
| LRG-SR-002 | No stored passwords | 5.1 / DS-SEC-01 | AUTH | TC-SEC-05 | N | |
| LRG-SR-003 | Server-side authorization | 5.1 / DS-SEC-01 | AUTH / RES | TC-AUTH-04, TC-SEC-06 | N | |
| LRG-SR-004 | Input validation / SQL | 5.1 / DS-SEC-01 | SRCH / RES | TC-SEC-02 | N | |
| LRG-SR-005 | Login lockout | 5.1 / DS-SEC-01 | AUTH | TC-AUTH-02 | N | |
| LRG-SR-006 | Cookie flags | 5.1 / DS-SEC-01 | AUTH / UI | TC-SEC-03 | N | |
| LRG-SR-007 | Rate limiting | 5.1 / DS-SEC-01 | SRCH / RES | TC-SEC-07 | N | |